Dev Gupta

Data Scientist · ML Engineer · Berlin, Germany

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M.Sc. Data Science, graduating February 2027 · Available full-time from March 2027

Summary

Data Science master's student with nearly 3 years of professional software engineering at NTT DATA. Builds end-to-end ML and LLM systems: training models in PyTorch, building local RAG applications, and serving them through FastAPI, Streamlit and Docker. Brings production habits from enterprise integration work — testing, CI/CD and secure APIs — to applied machine learning.

Experience

Software Engineer — NTT DATA, Bengaluru, India · 10/2021 – 07/2024

- Reduced integration delivery time by 30% by optimizing data pipelines and API workflows.
- Developed scalable REST APIs on the MuleSoft Anypoint Platform with RAML and DataWeave, connecting Salesforce, SAP and databases through API-led architecture.
- Secured APIs with OAuth 2.0 and JWT; contributed to CI/CD pipelines with Git, Jenkins and CloudHub.
- Monitored and debugged production applications in Runtime Manager, improving reliability and uptime.
- Designed end-to-end tests with MUnit and Postman, and wrote documentation used to onboard new developers.
- Recognized for excellence after MuleSoft training; led delivery of multiple high-impact integration projects in Agile/Scrum teams.

Projects

PaperPilot — Local-first AI research paper assistant · Python, Ollama, ChromaDB, Pydantic, Streamlit, Docker · GitHub

- Built an end-to-end RAG pipeline that answers questions only from an uploaded paper, with page-level citations, structured summaries and APA/MLA/IEEE/BibTeX export.
- Reduced hallucination risk by refusing low-relevance questions and validating every citation against retrieved sources; 220+ automated tests.

Pneumonia Detection from Chest X-rays · PyTorch, CNN, Grad-CAM, Streamlit · GitHub

- Built a custom CNN, handling ~2.9:1 class imbalance with weighted loss and a stratified validation split; analyzed sensitivity–specificity trade-offs across thresholds and explained predictions with Grad-CAM.

House Price Prediction — Full-stack ML app · PyTorch, FastAPI, Streamlit, Docker · GitHub

- Trained a neural network on ~21,600 King County sales and served it through a FastAPI REST API and Streamlit UI with Docker Compose, bundling scaler and feature order for consistent inference.

Euro Energy Data Pipeline · Apache Airflow, PostgreSQL, Docker, Streamlit · GitHub

- Built an Airflow pipeline ingesting Eurostat energy data for 8 countries (2018–2024) into PostgreSQL, with retries, validation, duplicate-safe inserts and an interactive dashboard.

Job Agent — LLM job-application assistant · Ollama (Llama 3.1 8B), FastAPI, Streamlit · GitHub

- Compares a CV with a job description using a local LLM and generates a match report, skill gaps, tailored cover letter and interview tips.

Education

M.Sc. Data Science — University of Europe for Applied Sciences, Berlin · 02/2026 – 02/2027 (expected)

- Planned thesis: Confidence Calibration in Quantized and Spiking Neural Networks (PyTorch, ResNet-20, CIFAR-10).

B.E. Computer Science and Engineering — Cambridge Institute of Technology (VTU), Bengaluru · 08/2018 – 08/2022

- Public Relations Lead, Google Developer Student Club.

Skills

- **Machine learning:** PyTorch, deep learning, CNNs, computer vision, class imbalance, model explainability (Grad-CAM)
- **LLMs and GenAI:** RAG, embeddings, ChromaDB, Ollama, structured LLM output (Pydantic), hallucination mitigation
- **Data:** Python, SQL, PostgreSQL, Apache Airflow, ETL, Streamlit dashboards
- **Engineering:** FastAPI, REST APIs, MuleSoft, Docker, Git, CI/CD (Jenkins), pytest, OAuth 2.0 / JWT

Languages

English (professional working proficiency) · Hindi (native)